

British Math Olympiad Rd 1 2021 — P2/6

Jonathan Kasongo

October 15, 2025

British Math Olympiad Rd 1 2021 — P2/6

One day Arun and Disha played several games of table tennis. At five points during the day, Arun calculated the percentage of the games played so far that he had won. The results of these calculations were exactly 30%, exactly 40%, exactly 50%, exactly 60% and exactly 70% in some order. What is the smallest possible number of games they played?

Solution

The smallest possible number of games is 30. Since $70\% = 7/10$ and $30\% = 3/10$ both of these percentages must occur after some $10k$ games have been played, where $k \in \mathbb{N}$. We observe three cases.

If the 30% occurs after exactly 10 games have been played, then $70\% = 14/20$ can't occur after exactly 20 games have been played since it would require Arun to win $14 - 3 = 11$ games whilst there are only 10 games between the 10th and 20th games.

If the $30\% = 6/20$ occurs after exactly 20 games have been played, then 70% can't occur after exactly 10 games have been played since it would require Arun to lose $7 - 6 = 1$ of his wins which is impossible.

If the 30% occurs after exactly $10k \geq 30$ games have been played then the number of games is definitely more than 30.

Thus 30 games is the minimum and it suffices to demonstrate a valid construction, we can have the order for example

$$50\% = \frac{1}{2}, \quad 60\% = \frac{3}{5}, \quad 30\% = \frac{3}{10}, \quad 40\% = \frac{6}{15}, \quad 70\% = \frac{21}{30}$$

which is valid since Arun needs to win some non-negative integer number of wins that is $\leq$ the number of games that separate the two percentage calculation points. ■